Akoizo Original Practice - 2027 Rotation Form 5

Division C | Astronomy | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Astronomical metals are…

Unit: Galaxies, stellar populations, and the interstellar medium.
Worked-case data: A region is dark in visible light but bright in infrared and contains embedded young sources. Explain without declaring it empty.

A. All solids only

B. Only charged atoms

C. Elements heavier than helium

D. Only iron and nickel

Answer: _____

2. (2 points) A giant can cool at the surface while brightening because…

Unit: Cloud collapse and mass-dependent stellar evolution.
Worked-case data: A toy main-sequence model uses L □ M^3·^5 and lifetime □ M/L. Compare a 2 Msun star with the Sun.

A. Temperature has no effect

B. Its area becomes zero

C. Its radius increases strongly

D. Distance must decrease

Answer: _____

3. (2 points) False-color blue necessarily means…

Unit: Observational astronomy and the electromagnetic spectrum.
Worked-case data: A star has parallax 50 mas and approximate blackbody temperature 5800 K.

A. A known temperature

B. A nearby object

C. Only the assigned display channel

D. Visible blue starlight

Answer: _____

4. (2 points) A flux-limited distant sample favors…

Unit: Advanced graph and image interpretation.
Worked-case data: Two stars have equal effective temperature but luminosities 1 and 100 Lsun. Compare radii.

A. Intrinsically luminous detectable objects

B. Every star equally

C. Only the lowest-mass stars

D. No selection effect

Answer: _____

5. (4 points) Calculate Rs for 10 Msun.

Unit: Supernovae, white dwarfs, neutron stars, and black holes.
Worked-case data: Use 3 km per solar mass as a rounded Schwarzschild-radius scale for a 10 Msun nonrotating black hole.

Answer: _____
_____
_____

6. (4 points) Why can a white dwarf be hot but faint?

Unit: Stellar properties, fusion, and the H-R diagram.
Worked-case data: A star has radius 2 Rsun and temperature 2 Tsun. Calculate L/Lsun.

Answer: _____
_____
_____

7. (4 points) Why not add magnitudes to combine stars?

Unit: Astrophysics equations and quantitative synthesis.
Worked-case data: A source has m = 10 and M = 5, with negligible extinction. Find distance and compare its flux with the same source at 10 pc.

Answer: _____
_____
_____

8. (4 points) Why not model a galaxy as one star?

Unit: Galaxies, stellar populations, and the interstellar medium.
Worked-case data: A region is dark in visible light but bright in infrared and contains embedded young sources. Explain without declaring it empty.

Answer: _____
_____
_____

9. (4 points) Name two turnoff assumptions or corrections.

Unit: Cloud collapse and mass-dependent stellar evolution.
Worked-case data: A toy main-sequence model uses L □ M^3·^5 and lifetime □ M/L. Compare a 2 Msun star with the Sun.

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: A star has parallax 50 mas and approximate blackbody temperature 5800 K.

Answer: _____
_____
_____

11. (2 points) A synthetic star has parallax 0.002 arcseconds. Use d(pc)=1/p to find distance.

Answer: _____
_____
_____

12. (2 points) A standard-candle exercise gives apparent magnitude 15, absolute magnitude 5 and negligible extinction. Use m-M=5log10(d/10 pc). Find d.

Answer: _____
_____
_____

13. (2 points) A star has twice the reference radius and the same temperature. Find its luminosity ratio using L proportional to R^2 T^4.

Answer: _____
_____
_____

14. (2 points) Use Wien's constant 2.90e-3 m K for a 5,800 K star. Find peak wavelength.

Answer: _____
_____
_____

15. (2 points) An artificial galaxy recession speed is 350 km/s. Use H0=70 km/s/Mpc and the nearby linear Hubble law to find distance.

Answer: _____
_____
_____

16. (2 points) A normalized binary exercise has separation 2 AU and period 2 years. Use total mass in solar masses=a^3/P^2. Find total mass.

Answer: _____
_____
_____

17. (2 points) NASA identifies M82 X-2 as an ultraluminous pulsar. What does a coherent rotation-linked pulse train imply about the compact object, and why is luminosity alone insufficient?

Answer: _____
_____
_____

18. (2 points) SN 2014J is a Type Ia supernova in M82. Explain why standardized Type Ia luminosities can constrain distance and why dust must be considered.

Answer: _____
_____
_____

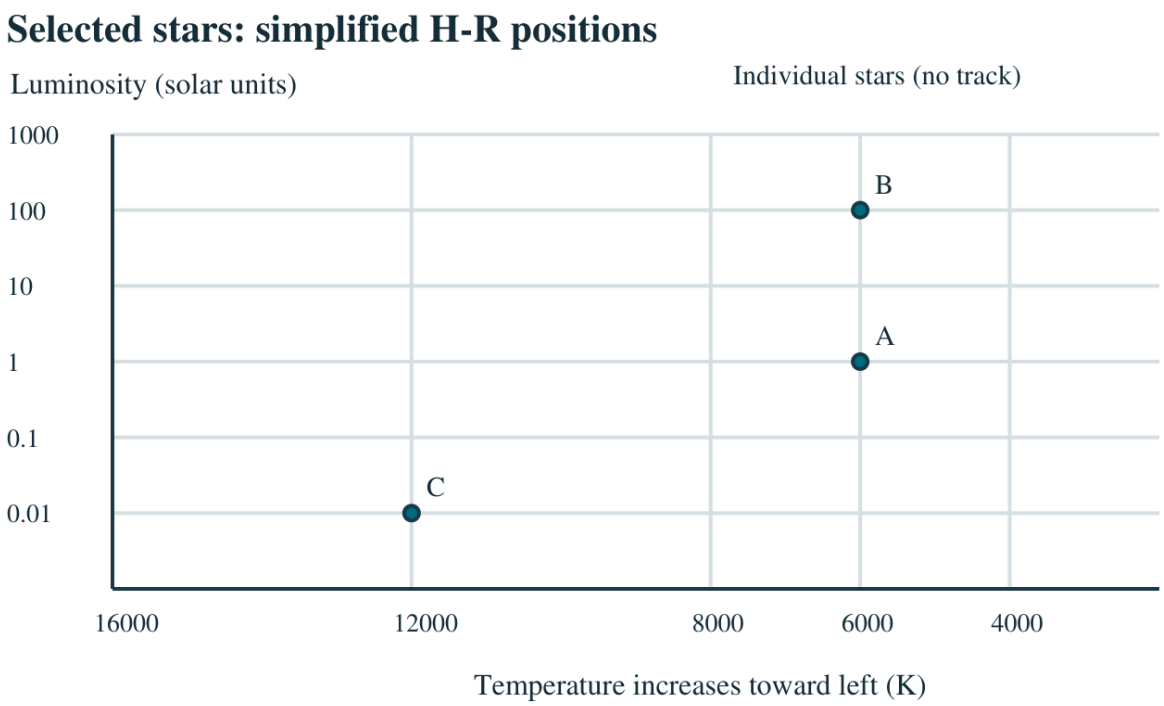
19. (2 points) GW170817 was observed in gravitational waves and electromagnetic radiation. Explain the value of this multi-messenger evidence for a neutron-star merger.

Answer: _____
_____
_____

20. (2 points) Synthetic starburst-galaxy case 5: ultraviolet emission and infrared dust emission both increase. Explain why using only visible light can miss part of the recent star formation.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Use L proportional to R^2 T^4 to find radius B / radius A.



F1. Hertzsprung-Russell diagram (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Which labeled object best fits a hot, low-luminosity compact remnant position? Explain.

Refer to F1. Hertzsprung-Russell diagram (original synthetic teaching figure) shown with the first question in this figure group.

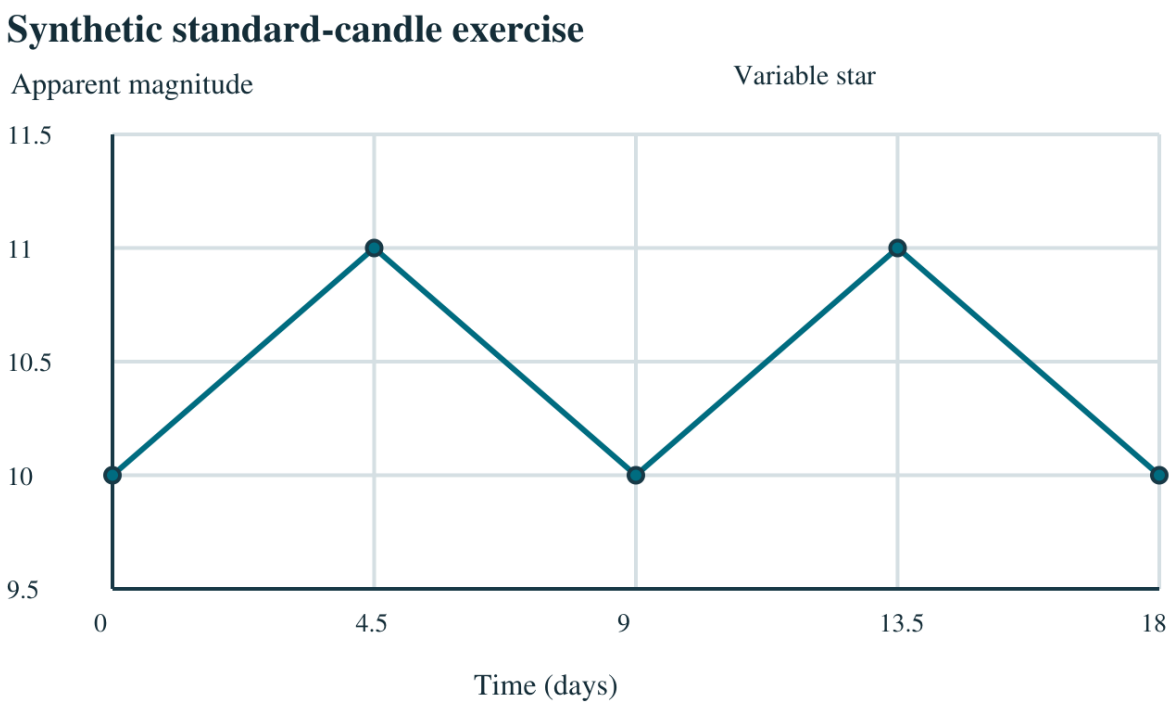
Answer: _____
_____
_____

23. (2 points) Use Figure F1. Why should the displayed positions not be read as motion of one star through space?

Refer to F1. Hertzsprung-Russell diagram (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Read the repeat period.



F2. Pulsating-variable photometry (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. For this teaching estimate use the arithmetic midpoint of the plotted magnitude extrema as m. Take M=0.5, negligible extinction, and m-M=5log10(d/10pc). Find distance.

Refer to F2. Pulsating-variable photometry (original synthetic teaching figure) shown with the first question in this figure group.

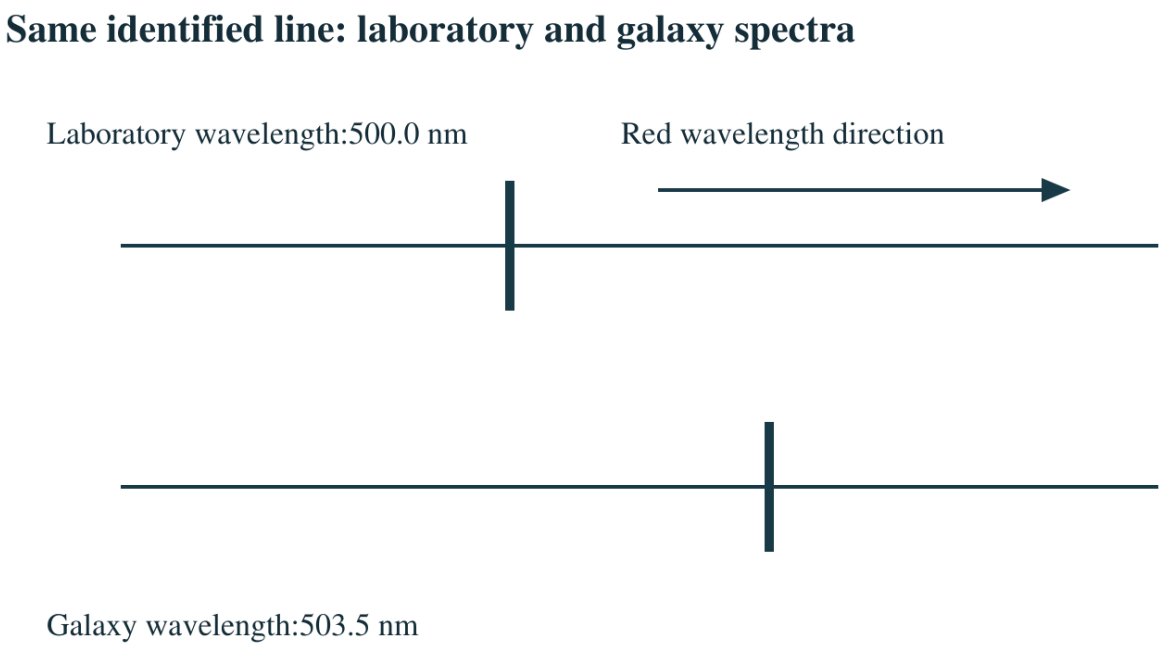
Answer: _____
_____
_____

26. (2 points) Use Figure F2. At plotted magnitude10 versus11, which state is brighter and why?

Refer to F2. Pulsating-variable photometry (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Calculate z=(observed-rest)/rest.



F3. Spectral-line displacement (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Use the low-redshift relation v=cz, with c=300000 km/s, to estimate recession speed.

Refer to F3. Spectral-line displacement (original synthetic teaching figure) shown with the first question in this figure group.

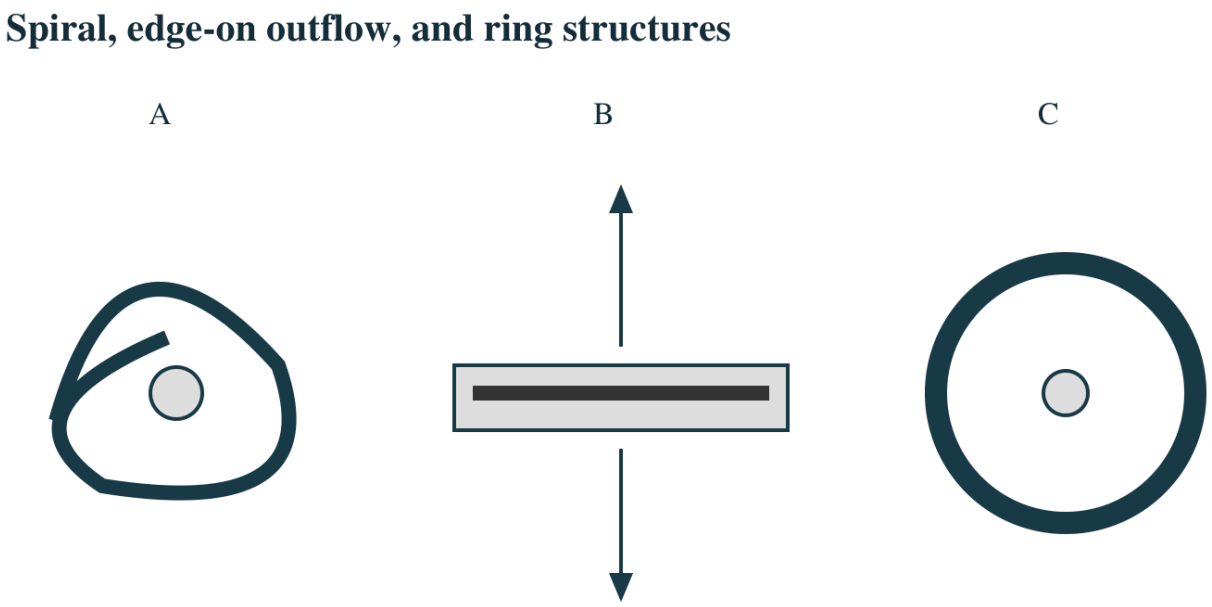
Answer: _____
_____
_____

29. (2 points) Use Figure F3. Use H0=70 km/s/Mpc and the nearby linear Hubble law to estimate distance.

Refer to F3. Spectral-line displacement (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Which panel visibly shows a ring morphology, relevant when studying the listed Cartwheel Galaxy?



Original morphology sketches; structures are schematic, not telescope images.

F4. Galaxy morphology comparison (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Which panel illustrates an outflow perpendicular to an edge-on disk, a geometry relevant to starburst-galaxy studies?

Refer to F4. Galaxy morphology comparison (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. Why cannot these three morphology sketches alone establish a universal galaxy age sequence?

Refer to F4. Galaxy morphology comparison (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____